

Programme	B. Sc. Mathematics Honours				
Course Code	MAT5EJ307				
Course Title	Foundations of Mathematics				
Type of Course	Major				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Fundamental Mathematics Concepts: Set, Functions, Logic				
Course Summary	This course exposes the set theoretic foundations of mathematics, exposes the problems and challenges that it is wrought with, and briefly introduces rigorous ways of tackling the same.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the broad, intuitive foundations on which all of modern mathematics is based	U	C	Internal Examination/Seminar/ Assignment/ End Sem examination
CO2	Learn a reasonably rigorous, though informal way of defining and establishing properties of transfinite arithmetic –cardinals and ordinals	Ap	P	Internal Examination/Seminar/ Assignment/ End Sem examination
CO3	Understand and appreciate the paradoxes that arise from the informal foundations that are taken for granted.	An	C	Internal Examination/Seminar/ Assignment/ End Sem examination
CO4	Understand how a consistent framework of set theory could be attempted by modifying, limiting, and extending the intuitive framework.	C	M	Internal Examination/Seminar/ Assignment/ End Sem examination

CO5	Appreciate the role of rigour and formal symbolic logic in clarifying and establishing mathematical foundations.	U	C	Internal Examination/Seminar/ Assignment/ End Sem examination
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook		Set Theory & Logic, Robert R. Stoll, 1979, Dover Pub. Inc., ISBN-10 : 0486638294, ISBN-13: 978-0486638294		
Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Naive Set Theory :Chapter 1: Sets & Relations		11	25
	1	§1,2: Cantor's Concept of a Set, The Basis of Intuitive Set Theory	2	
	2	§3,4,5: Inclusion, Operations for Sets, The Algebra of Sets	2	
	3	§6, §7, §8: Relations, Equivalence Relations, Functions	1	
	4	§9: Composition & Inversion for Functions	1	
	5	§10: Operations for Collections of Sets	1	
	6	§11: Ordering Relations – POSETs, TOSETs, WOSETs	2	
II	Cardinal & Ordinal Numbers: Chapter 2: The Natural Number Sequence & Its Generalisations		20	25
	7	§1: The Natural Number Sequence	3	
	8	§2: Proof & Definition by Induction	2	
	9	§3: Cardinal Numbers	2	
	10	§4: Countable Sets	2	
	11	§5: Cardinal Arithmetic	2	
	12	§6, §7: Order Types, Well Ordered Sets & Ordinal Numbers	1	

	13	§8: Axiom of Choice, Well Ordering Theorem & Zorn's Lemma (Proof of the technical Theorem 8.1 relegated to OpenEnded Module)	2	
	14	§9, §11,: Further Properties of Cardinal Numbers, The Paradoxes of Intuitive Set Theory	2	
III	Mathematical Logic: Chapter 4: Logic Chapter 5: Informal Axiomatic Mathematics		9	10
	15	§1-5: Statement Calculus – Connectives, Truth Tables, Validity, Consequence, Applications	3	
	16	§6-9: Predicate Calculus – Formulation, Validity, Consequence, Applications	3	
	17	Chapter 5: Informal Axiomatic Mathematics – A Quick Review of Chapter 5	3	
IV	Axiomatic Set Theory Chapter 7: Informal Axiomatic Set Theory		8	10
	18	§1, §2, §3: The Axioms of Extension & Set Formation, Pairing, Union & Power Set	1	
	19	§4, §5, §6: The Axiom of Infinity, The Axiom of Choice, The Axiom Schemas of Replacement & Restriction	1	
	20	§7: Ordinal Numbers	1	
	21	§8: Ordinal Arithmetic	1	
	22	§9: Cardinal Numbers & Their Arithmetic	1	
V	Open Ended Module		12	
	1	Chap 2 Section 10: Some Theorems Equivalent to Axiom of Choice	3	
	2	Chap 2 Proof of Theorem 8.1	3	
	3	Chap 3: The Extension of the Natural Numbers to the Real Numbers	3	
	4	Chap 7 Section 10: The von Neumann-Bernays-Godel Theory of Sets	3	

References:

1. Naive Set Theory, Paul R. Halmos, Dover (2015).
2. Axiomatic Set Theory, Patrick Suppes, 1972, Dover.
3. A Tour Through Mathematical Logic, Robert Wolff, 2005, MAA Press.
4. Set Theory, Thomas Jech, Springer (2003)

Note: The 70 marks shown in the last column, distributed over the first four modules, is only for the external examination.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	1	3	3	0	1	0	3	1	2
CO 2	3	3	2	3	3	0	1	0	3	1	2
CO 3	3	0	0	3	3	0	1	0	3	1	2
CO 4	3	2	2	3	3	0	1	0	3	1	2
CO 5	3	1	1	3	3	0	1	0	3	1	2

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓
CO 4	✓	✓	✓	✓	✓
CO 5	✓	✓	✓	✓	✓